

GOKUL G. ANUGRAH

Lexington, KY • (859) 285-1611 • gokulanugrah1@gmail.com • linkedin.com/in/gokul-anugrah-20133518
No Visa Sponsorship Required

RESEARCH INTERESTS

Cardiovascular computational mechanics • Fluid-structure interaction (FSI) • High-fidelity CFD/DNS of hemodynamic flows • Machine learning surrogate models for computational simulations • HPC-based solver development

EDUCATION

Ph.D., Mechanical Engineering

December 2025

University of Kentucky, Lexington, KY

Dissertation: High-fidelity CFD: DNS of patient-specific aortic flows, FSI of parachute deployment in supersonic flow, porous media interface modeling

M.Sc., Computational Mechanics

January 2016

University of Duisburg-Essen, Germany (Master's thesis conducted at DLR Göttingen)

Coursework: CFD, Numerical Mathematics, FEM, Continuum Mechanics, Turbulent Flows, C++/Fortran

B.Tech., Mechanical Engineering

April 2011

National Institute of Technology, Warangal, India

Coursework: CFD, Heat Transfer, FEM, Numerical Methods, C++

TECHNICAL SKILLS

Cardiovascular / Biomedical Simulation: DNS of patient-specific aortic dissection geometries; pulsatile flow; immersed boundary method (IBM); Pointwise for patient-specific meshing; 3D Slicer familiarity for clinical geometry processing

Computational Methods (FEA/CFD/FSI): Finite Element Analysis (FEA); Fluid-Structure Interaction (FSI); steady & unsteady RANS, LES, DES/DES, DNS; compressible/supersonic flows; conjugate heat transfer; multiphase flows; porous media modeling

CFD Software: OpenFOAM, ANSYS Fluent, STAR-CCM+, CFD++ (IIST), TAU Code (DLR); in-house Fortran CFD solver and C++ structural solver (University of Kentucky)

Machine Learning & Surrogate Modeling: ML-based surrogate modeling (personal projects); Python/PyTorch; physics-informed neural networks (PINNs) exposure; ML-CFD hybrid workflow integration; certified: Machine Learning (Stanford/Coursera), Data Science in Python (U. Michigan/Coursera)

Mesh Generation: Pointwise (unstructured/hybrid); STAR-CCM+ (surface & volume meshing); complex geometry clean-up; upto ~100M-cell IBM meshes for DNS

Programming: Python (automation, scientific computing), C/C++, Fortran (solver development), MATLAB

HPC & Infrastructure: MPI-based parallel computing, Linux/HPC cluster environments (10 years), massively parallel simulations, Git

Post-Processing & Validation: ParaView, Tecplot, VisIt; Python-based batch post-processing; CFD validation against experimental and clinical reference data; mesh sensitivity studies.

RESEARCH EXPERIENCE

Research Assistant / Teaching Assistant

October 2019 – December 2025

University of Kentucky | Lexington, KY

Cardiovascular Computational Mechanics:

- Conducted DNS of patient-specific aortic dissection geometries to characterize unsteady turbulent hemodynamic behavior; validated solver against published clinical reference data (manuscript under review, Physics of Fluids).
- Generated production-quality patient-specific aortic surface mesh (~100K points) from clinical geometry using 3D Slicer and Pointwise; constructed ~56M-cell IBM volume mesh for high-fidelity DNS; performed systematic mesh sensitivity studies.

- Simulated turbulence transition in pulsatile curved-tube flows relevant to vascular hemodynamics (manuscript under review, Physics of Fluids).

Fluid-Structure Interaction & Solver Development:

- Developed and maintained an in-house Fortran-based compressible CFD solver coupled with a C++ structural solver for FSI simulations of supersonic parachute deployment on HPC clusters — demonstrating end-to-end ownership of coupled FSI simulation tools.
- Implemented a porous media interface boundary condition in the in-house Fortran CFD solver for supersonic broadcloth parachute simulations (published: Boustani et al., AIAA 2020).

Machine Learning & Workflow Automation:

- Conducted personal surrogate modeling projects using ML techniques for accelerating CFD parameter studies.
- Applied OpenFOAM for additional cardiovascular and internal flow simulations through independent study, broadening solver expertise beyond primary research platforms.

Teaching & Mentoring:

- Delivered undergraduate lectures in fluid mechanics, heat transfer, and thermodynamics; mentored junior researchers in CFD methodology, HPC workflows, and technical communication.

Senior Project Fellow

January 2018 – September 2019

Indian Institute of Space Science and Technology (IIST) | Thiruvananthapuram, India

- Simulated jet-in-crossflow in a rocket nozzle using CFD++ and OpenFOAM under supersonic conditions; validated thrust augmentation results against free-jet experimental data.
- Performed conjugate heat transfer (CHT) simulations in OpenFOAM for propulsion thermal management; developed Fortran code for supersonic reactive flow simulations.
- Authored Python pre/post-processing automation scripts; published 1 peer-reviewed journal article and 1 conference paper.

Research Assistant (Master's Thesis)

April 2015 – January 2016

German Aerospace Center (DLR) — Institute of Aerodynamics & Flow Technology | Göttingen, Germany

- Conducted steady and unsteady turbulent flow simulations using RANS, LES, DES, and DDES in DLR's TAU Code (unstructured solver) on complex aerospace geometries.
- Contributed to solver development and rigorous validation benchmarking across the RANS-to-scale-resolving turbulence modeling spectrum.

PUBLICATIONS & PRESENTATIONS

Journal Articles:

- Anugrah, G.G., et al. Turbulence and Transition in Pulsating Curved-Tube Flows. Under review, Physics of Fluids.
- Anugrah, G.G., et al. High-Fidelity Simulation of Patient-Specific Aortic Dissection. Under Review, Journal of Biomechanical Engineering.
- Anugrah, G.G., et al. (2019). Secondary Injection in Nozzle Divergence Section for Thrust Augmentation. Journal of Applied Fluid Mechanics. DOI: 10.29252/JAFM.12.05.29842.

Conference Proceedings:

- Boustani, J., Anugrah, G.G., et al. (2020). Numerical Investigation of Parachute Deployment in Supersonic Flow. AIAA SciTech. DOI: 10.2514/6.2020-1050.
- Presentations at Summer Biomechanics, Bioengineering & Biotransport Conference (SB3C 2023, Vail, CO) and WCCM-PANACM 2024 (Vancouver, BC).
- "Numerical Studies on Thrust Augmentation in Nuclear Thermal Rocket by Secondary Injection." National Seminar on Aerospace Propulsion, India: Best Paper Award.

CERTIFICATIONS & OUTREACH

- Machine Learning — Stanford University / Coursera
- Data Science in Python — University of Michigan / Coursera
- SimScale Student Ambassador — Introduced cloud-based CFD tools to 100+ students (2018); developed curriculum and delivered hands-on training.